Experiment to understand the data preprocessing in Data science

In [2]:
```
import numpy as np
import pandas as pd
```

In [34]:
```
df = pd.read_excel("Hotel_Dataset.xlsx")
df
```

Out[34]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax
0	1	20-25	4	Ibis	veg	1300	2
1	2	30-35	5	LemonTree	Non-Veg	2000	3
2	3	25-30	6	RedFox	Veg	1322	2
3	4	20-25	-1	LemonTree	Veg	1234	2
4	5	35+	3	Ibis	Vegetarian	989	2
5	6	35+	3	Ibys	Non-Veg	1909	2
6	7	35+	4	RedFox	Vegetarian	1000	-1
7	8	20-25	7	LemonTree	Veg	2999	-10
8	9	25-30	2	Ibis	Non-Veg	3456	3
9	9	25-30	2	Ibis	Non-Veg	3456	3
10	10	30-35	5	RedFox	non-Veg	-6755	4

In [35]:
```
df.duplicated()
```

Out[35]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool
```

In [5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     str
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     str
 4   FoodPreference   11 non-null     str
 5   Bill             11 non-null     int64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group.1      11 non-null     str
dtypes: int64(5), str(4)
memory usage: 924.0 bytes
```

In [6]:
```
df.drop_duplicates(inplace=True)
df
```

Out[6]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax
0	1	20-25	4	Ibis	veg	1300	2
1	2	30-35	5	LemonTree	Non-Veg	2000	3
2	3	25-30	6	RedFox	Veg	1322	2
3	4	20-25	-1	LemonTree	Veg	1234	2
4	5	35+	3	Ibis	Vegetarian	989	2
5	6	35+	3	Ibys	Non-Veg	1909	2
6	7	35+	4	RedFox	Vegetarian	1000	-1
7	8	20-25	7	LemonTree	Veg	2999	-10
8	9	25-30	2	Ibis	Non-Veg	3456	3
10	10	30-35	5	RedFox	non-Veg	-6755	4

In [7]:
```
len(df)
```

Out[7]:
```
10
```

In [8]:
```
index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
index
```

Out[8]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [9]:
```
df
```

Out[9]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	E
0	1	20-25	4	Ibis	veg	1300	2	
1	2	30-35	5	LemonTree	Non-Veg	2000	3	
2	3	25-30	6	RedFox	Veg	1322	2	
3	4	20-25	-1	LemonTree	Veg	1234	2	
4	5	35+	3	Ibis	Vegetarian	989	2	
5	6	35+	3	Ibys	Non-Veg	1909	2	
6	7	35+	4	RedFox	Vegetarian	1000	-1	
7	8	20-25	7	LemonTree	Veg	2999	-10	
8	9	25-30	2	Ibis	Non-Veg	3456	3	
9	10	30-35	5	RedFox	non-Veg	-6755	4	

In [10]:
```
df.drop(['Age_Group.1'], axis=1, inplace=True)
df
```

Out[10]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	E
0	1	20-25	4	Ibis	veg	1300	2	
1	2	30-35	5	LemonTree	Non-Veg	2000	3	
2	3	25-30	6	RedFox	Veg	1322	2	
3	4	20-25	-1	LemonTree	Veg	1234	2	
4	5	35+	3	Ibis	Vegetarian	989	2	
5	6	35+	3	Ibys	Non-Veg	1909	2	
6	7	35+	4	RedFox	Vegetarian	1000	-1	
7	8	20-25	7	LemonTree	Veg	2999	-10	
8	9	25-30	2	Ibis	Non-Veg	3456	3	
9	10	30-35	5	RedFox	non-Veg	-6755	4	

In [12]:
```
df.loc[df['CustomerID'] < 0, 'CustomerID'] = np.nan
df.loc[df['Bill'] < 0, 'Bill'] = np.nan
df.loc[df['EstimatedSalary'] < 0, 'EstimatedSalary'] = np.nan

df
```

Out[12]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax
0	1.0	20-25	4	Ibis	veg	1300.0	2
1	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3
2	3.0	25-30	6	RedFox	Veg	1322.0	2
3	4.0	20-25	-1	LemonTree	Veg	1234.0	2
4	5.0	35+	3	Ibis	Vegetarian	989.0	2
5	6.0	35+	3	Ibys	Non-Veg	1909.0	2
6	7.0	35+	4	RedFox	Vegetarian	1000.0	-1
7	8.0	20-25	7	LemonTree	Veg	2999.0	-10
8	9.0	25-30	2	Ibis	Non-Veg	3456.0	3
9	10.0	30-35	5	RedFox	non-Veg	NaN	4

In [13]:
```
df.loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20), 'NoOfPax'] = np.nan
df
```

Out[13]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax
0	1.0	20-25	4	Ibis	veg	1300.0	2.0
1	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3.0
2	3.0	25-30	6	RedFox	Veg	1322.0	2.0
3	4.0	20-25	-1	LemonTree	Veg	1234.0	2.0
4	5.0	35+	3	Ibis	Vegetarian	989.0	2.0
5	6.0	35+	3	Ibys	Non-Veg	1909.0	2.0
6	7.0	35+	4	RedFox	Vegetarian	1000.0	NaN
7	8.0	20-25	7	LemonTree	Veg	2999.0	NaN
8	9.0	25-30	2	Ibis	Non-Veg	3456.0	3.0
9	10.0	30-35	5	RedFox	non-Veg	NaN	4.0

In [14]:
```
df.Age_Group.unique()
```

Out[14]:
```
<StringArray>
['20-25', '30-35', '25-30', '35+']
Length: 4, dtype: str
```

In [15]:
```
df.Hotel.unique()
```

Out[15]:
```
<StringArray>
['Ibis', 'LemonTree', 'RedFox', 'Ibys']
Length: 4, dtype: str
```

In [30]:
```
df['Hotel'] = df['Hotel'].replace('Ibys', 'Ibis')
```

In [18]:
```
df.FoodPreference.unique()
```

Out[18]:
```
<StringArray>
['veg', 'Non-Veg', 'Veg', 'Vegetarian', 'non-Veg']
Length: 5, dtype: str
```

In [19]:
```
df.FoodPreference
```

Out[19]:
```
0           veg
1       Non-Veg
2           Veg
3           Veg
4    Vegetarian
5       Non-Veg
6    Vegetarian
7           Veg
8       Non-Veg
9       non-Veg
Name: FoodPreference, dtype: str
```

In [32]:
```
df['FoodPreference'] = df['FoodPreference'].replace(
    {
        'Vegetarian': 'Veg',
        'veg': 'Veg',
        'non-Veg': 'Non-Veg'
    }
)
```

In [23]:
```
df['EstimatedSalary'] = df['EstimatedSalary'].fillna(
    round(df['EstimatedSalary'].mean())
)
```

In [26]:
```
df['NoOfPax'] = df['NoOfPax'].fillna(
    round(df['NoOfPax'].median())
)
```

In [27]:
```
df['Rating(1-5)'] = df['Rating(1-5)'].fillna(
    round(df['Rating(1-5)'].median())
)
```

In [28]:
```
df['Bill'] = df['Bill'].fillna(
    round(df['Bill'].mean())
)
```

In [33]:
```
df
```

Out[33]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax
0	1.0	20-25	4	Ibis	Veg	1300.0	2.0
1	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3.0
2	3.0	25-30	6	RedFox	Veg	1322.0	2.0
3	4.0	20-25	-1	LemonTree	Veg	1234.0	2.0
4	5.0	35+	3	Ibis	Veg	989.0	2.0
5	6.0	35+	3	Ibis	Non-Veg	1909.0	2.0
6	7.0	35+	4	RedFox	Veg	1000.0	2.0
7	8.0	20-25	7	LemonTree	Veg	2999.0	2.0
8	9.0	25-30	2	Ibis	Non-Veg	3456.0	3.0
9	10.0	30-35	5	RedFox	Non-Veg	1801.0	4.0

In [ ]: